

APEX LOW VOLTAGE & SECURITY SERVICES

RS485 Access Control Wiring Map — Kantech KT Controller Daisy-Chain Reference

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Document Type	Site As-Built / Field Wiring Reference (non-SOP)
Site	[SITE NAME — confirm with Paul before distribution]
System	Kantech Access Control — RS485 Controller Bus
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Purpose: This document records the physical location and RS485 daisy-chain wiring sequence of all Kantech (KT) access control controllers on site. It is intended as a field reference for technicians performing service, troubleshooting, or expansion work on the access control system. It is not a safety procedure and does not replace SOP-LV series documents governing field work practices.

Lower Level KT Locations

Room 017b — Electrical Room

KT Device	Notes / Function
KT-001	MASTER KT — IP connection to head end
KT-013	—
KT-027	—
KT-002	RS485 bus continues to Room 038

Room 038 — Electrical Room (off Boiler Room)

Bus fed from KT-002 in Room 017b.

KT Device	Notes / Function
KT-003	Fed from KT-002 (Room 017b)
KT-004	—
KT-005	Also carries the spliced CAT feed to KT-009 (7th Fire LAN closet) — see Room 111 note below
KT-006	RS485 bus continues to Room 040

Room 040 — LAN Room

Bus fed from KT-002.

KT Device	Notes / Function
KT-014	Fed from KT-002
KT-015	—
KT-016	—
KT-022	RS485 bus continues to Room 020

Room 020 — Mechanical Room

KT Device	Notes / Function
KT-011	—
KT-012	RS485 bus continues to Room 111 (Upper Level)

Upper Level KT Locations

Room 111 — Mechanical Room

KT Device	Notes / Function
KT-017	—
KT-018	—
KT-019	—
KT-020	—
KT-021	—
KT-026	RS485 bus returns to KT-005 in Room 038, then splices onto CAT cable to feed the 7th Fire / LAN closet at KT-009

Room 145 (Boardroom) — Main Floor Board / LAN Closet ("7th Fire")

Access to this KT group is via Room 145 (Boardroom), Main Floor Board, into the LAN closet.

KT Device	Notes / Function
KT-007	—
KT-008	—
KT-009	RS485 feed enters here from KT-005 in Room 038 (spliced CAT run via KT-026, Room 111)
KT-010	—
KT-023	—
KT-024	—
KT-025	—
KT-028	—

Bus Sequence Summary

Full daisy-chain path, head end to end of line:

KT-001 (Master/IP, Rm 017b) → KT-013 → KT-027 → KT-002 → [Rm 038] KT-003 → KT-004 → KT-005 → KT-006 → [Rm 040] KT-014 → KT-015 → KT-016 → KT-022 → [Rm 020] KT-011 → KT-012 → [Rm 111] KT-017 → KT-018 → KT-019 → KT-020 → KT-021 → KT-026 → splice back to KT-005 (Rm 038) → CAT feed to 7th Fire LAN closet → [Rm 145 access] KT-009 → KT-007 → KT-008 → KT-010 → KT-023 → KT-024 → KT-025 → KT-028.

Field Notes

- KT-001 is the master controller with the IP connection to the head end server — all troubleshooting of the RS485 bus should start here.
- The splice at KT-026 (Room 111) back to KT-005 (Room 038), then onto CAT cable feeding KT-009, is the single point connecting the Upper Level Room 111 branch to the Room 145 / 7th Fire LAN closet branch. Treat this splice as a priority troubleshooting point for any bus fault affecting Room 145 and beyond.
- Room 145 (Boardroom) access is required to reach the Main Floor Board and the 7th Fire LAN closet KTs (KT-007 through KT-010, KT-023 through KT-025, KT-028).
- Confirm door/room numbers against current signage before performing work — room numbering may not match historical naming used in prior wiring diagrams.

End of document.